

Machine Learning

Unsupervised Learning

FAST

DISCOVERING
THE FUTURE

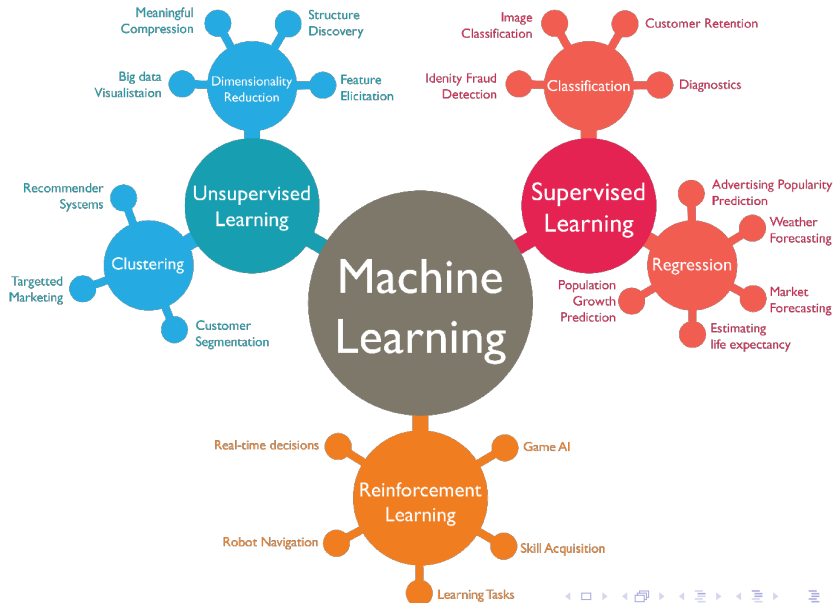
Topics of previous lectures

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics, Basic Linear Classifier
- ✓ K-Nearest Neighbours and Naive Bayes Classifier
- ✓ Linear and Quadratic Discriminant Analysis
- ✓ Support Vector Machines (SVM)
- ✓ Decision Trees
- ✓ Ensemble Methods (Bagging, Weighted Voting, Stacking)
- ✓ Regression Methods
- ✓ Evaluation and Scoring of Classifiers
- ✓ Ensemble Methods (Boosting)

Topics of today's lecture

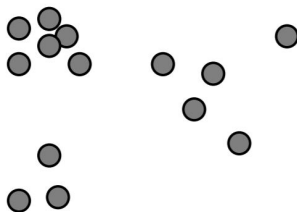
- Concepts of Cluster Analysis
- Hierarchical Clustering
- K-means Clustering
- K-medoids Clustering
- DBSCAN
- Dissimilarity Measure Selection

Machine Learning Map



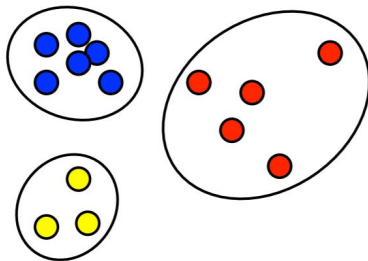
What is clustering?

- Given data objects



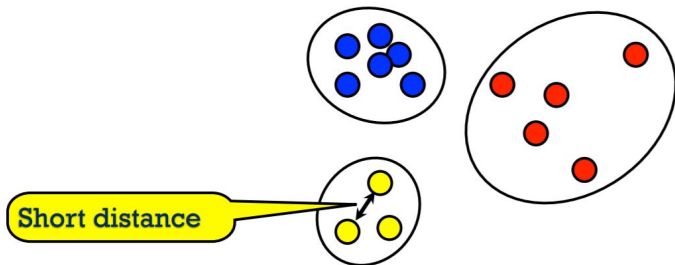
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- Find a grouping (**clustering**) such that the objects are:



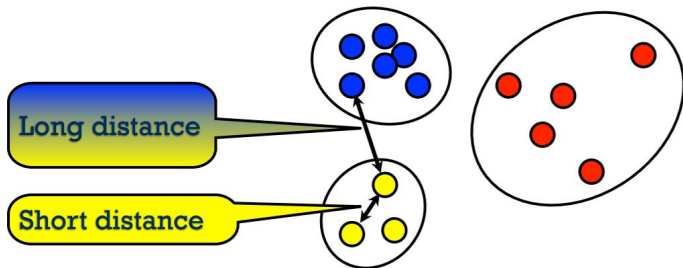
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What is clustering?

- Given data objects
- Find a grouping (**clustering**) such that the objects are:
 - similar (related) to the objects in the same group
 - dissimilar (unrelated) from objects in other groups



Why to cluster data?

- Intuition building

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- Hypothesis generation

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- Discover structures and patterns in high-dimensional data

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- Intuition building
- Hypothesis generation
- Discover structures and patterns in high-dimensional data
- Summarizing / compressing large data

Clustering is subjective

- Suppose that we need to put you in some groups for projects

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- How to define those groups?
 - Work experience
 - Age
 - Education
 - Preferences

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- How to define those groups?
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 - Preferences
- Which way is correct? Depends on the goal!

Examples

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- Image segmentation
- Search result grouping
- Social network analysis
- Anomaly detection

Example: Supermarket data

Customer 1



Task
Cluster customers into segments by their typical shopping baskets

Customer 3



Customer 4



Example: Supermarket data

Customer 1



Task

- Cluster customers into segments by their typical shopping baskets

Customer 3

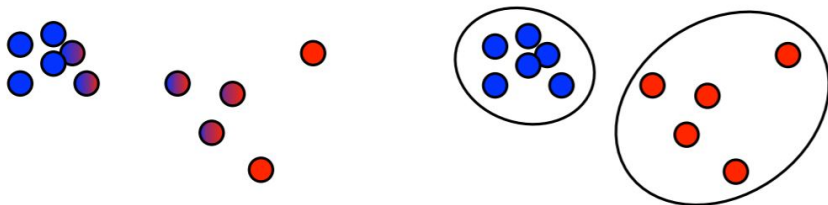


Motivation

- Launch a discount campaign addressing a particular segment of customers

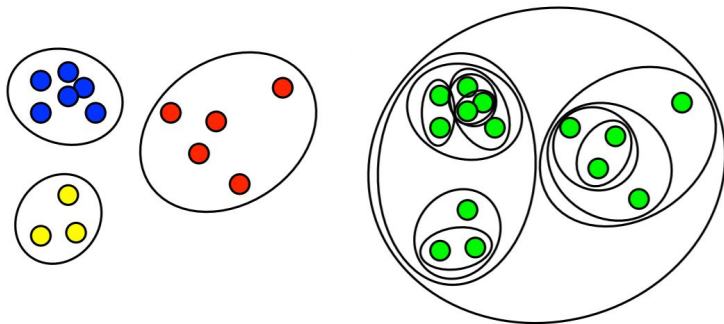
Clustering Types

- **Fuzzy (Soft) clustering:** Each object belongs to each cluster with some weight (the weight can be zero)
- **Non-fuzzy (Hard) clustering** each object belongs to exactly one cluster



Clustering Types

- **Partitional clustering** finds a fixed number of clusters
- **Hierarchical clustering** creates a series of clusterings contained in each other (nested clusters)



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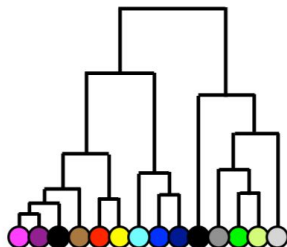
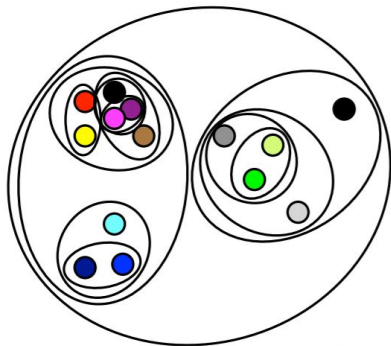
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 - **Agglomerative** clustering (Bottom-up): Start with a single-object clusters (singletons) and recursively merge them into larger clusters.
 - **Divisive** clustering (Top down): Start with a cluster containing all data points and recursively divide it into smaller clusters.

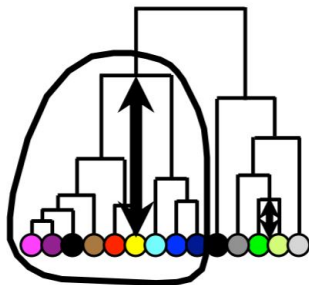
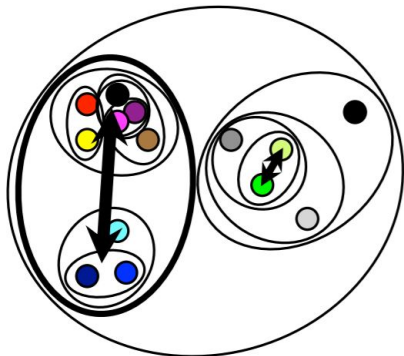
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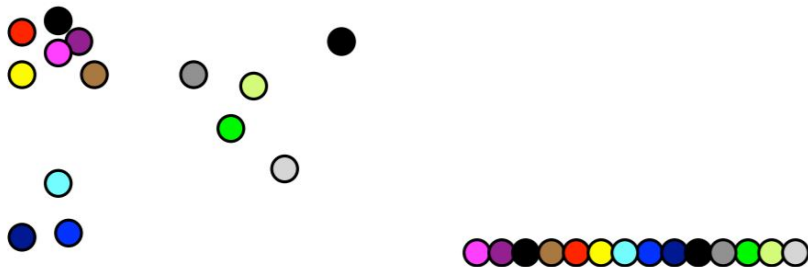
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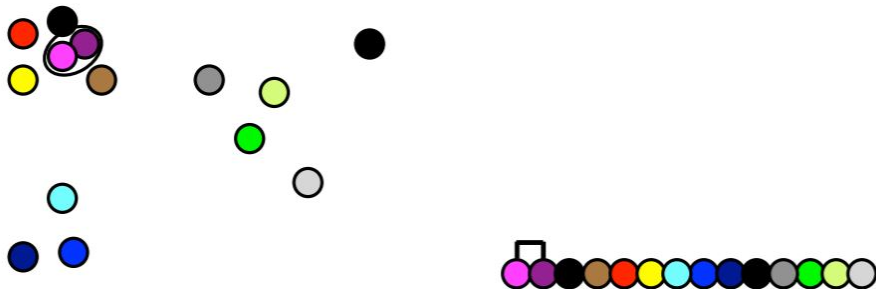
Hierarchical Clustering (Step 0)

Algorithm for Agglomerative Hierarchical Clustering: Join the two closest objects



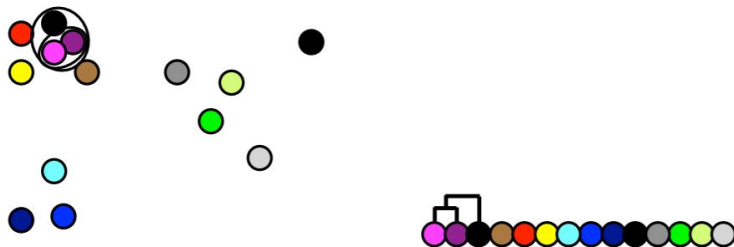
Hierarchical Clustering (Step 1)

Join the two closest objects (for example, closest with respect to Euclidean distance)



Hierarchical Clustering (Step 2)

Keep joining the closest pairs



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Keep joining the closest pairs

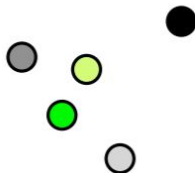


We will see how to measure distance between an individual item and a cluster!



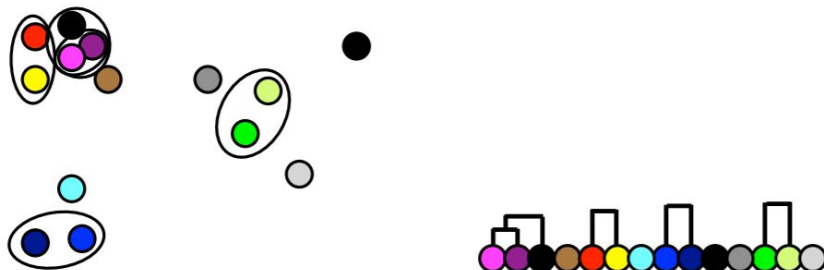
Hierarchical Clustering (Step 3)

Keep joining the closest pairs



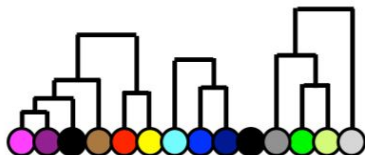
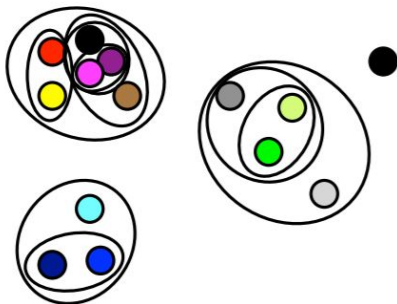
Hierarchical Clustering (Step 5)

Keep joining the closest pairs



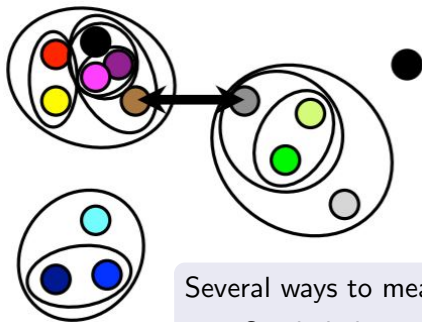
Hierarchical Clustering (Step 10)

After 10 steps we have 4 clusters left



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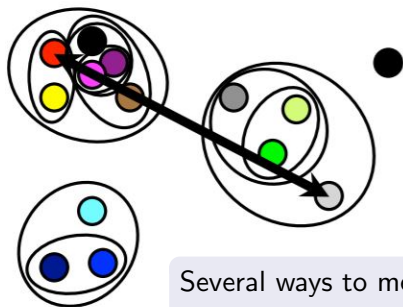


Several ways to measure distance between clusters:

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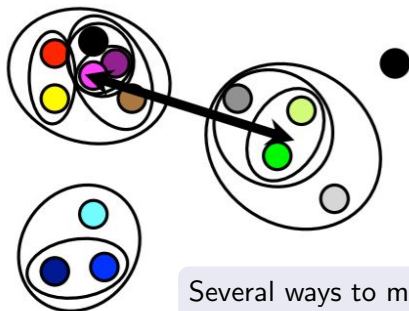


Several ways to measure distance between clusters:

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- Complete linkage (a.k.a furthest neighbour technique)

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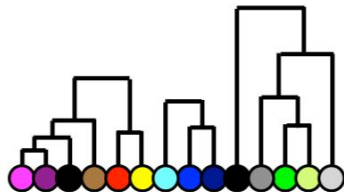
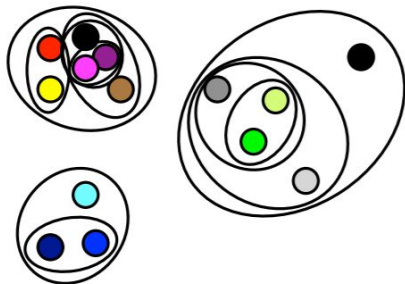


Several ways to measure distance between clusters:

- Single linkage (a.k.a. nearest neighbour technique)
- Complete linkage (a.k.a furthest neighbour technique)
- Average linkage
 - Weighted
 - Unweighted

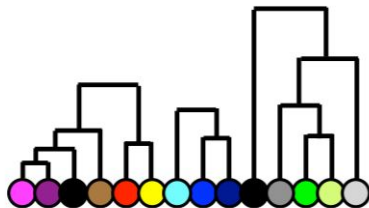
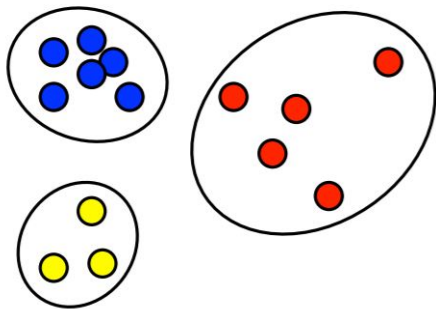
Hierarchical Clustering (Step 11)

In this example and at this stage we have the same result as in our partitional clustering example

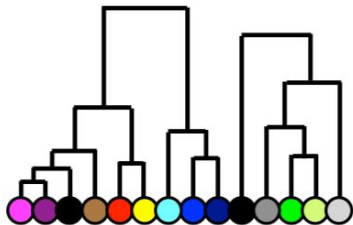
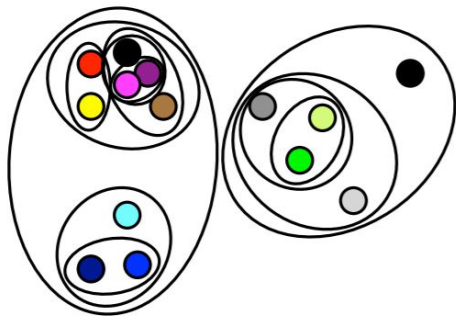


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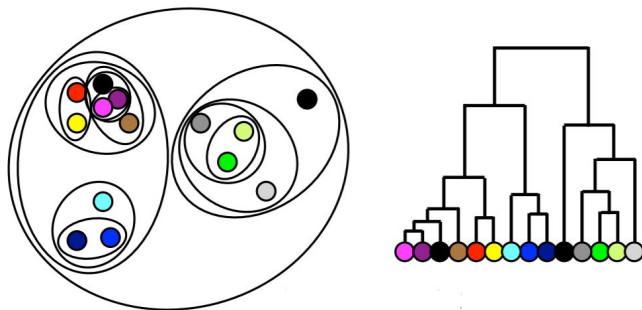


Hierarchical Clustering (Step 12)



Hierarchical Clustering (Step 13)

In the final step the two remaining clusters are joined into a single cluster



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 - Observations assigned to a cluster can be much closer to members of other clusters than they are to some members of their own cluster
- **Average** Linkage - distance is computed between clusters' centroids. This is a balanced approach: clusters tend to be relatively compact and relatively far apart.

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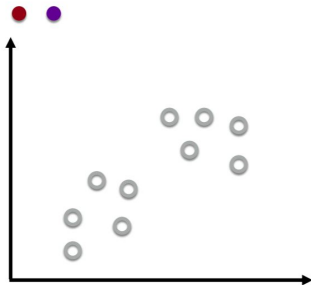
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- Pro:** In some problems, hierarchy of clusters is preferred over flat clusters
- Con:** Hierarchical clustering can't handle big data well, because of its quadratic time complexity

K-means Clustering

1. Choose K , the number of potential clusters

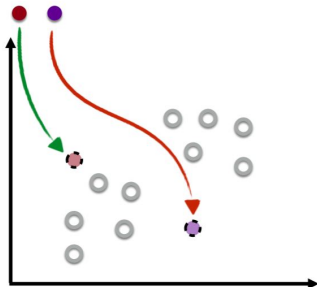
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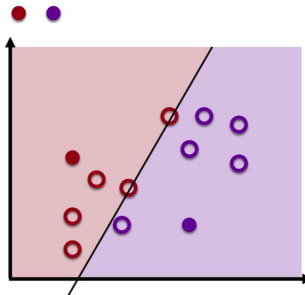
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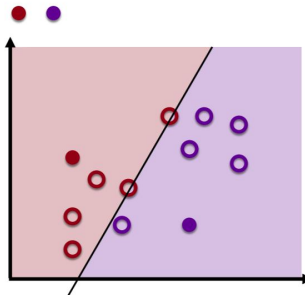
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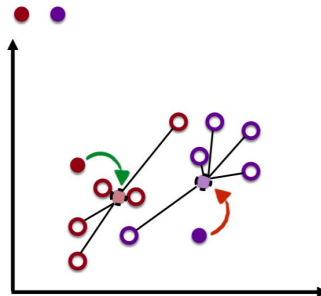
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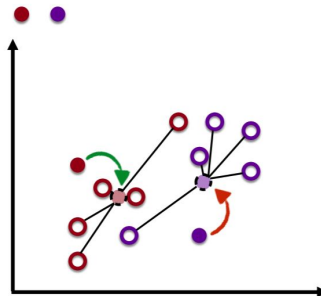
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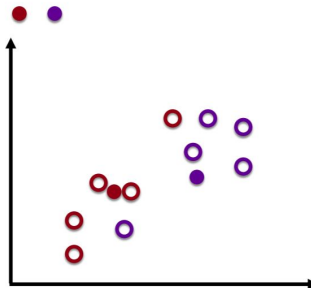
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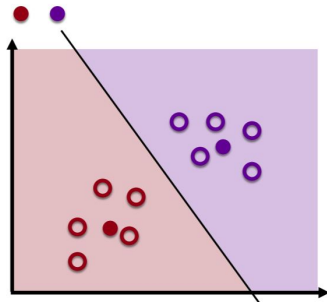
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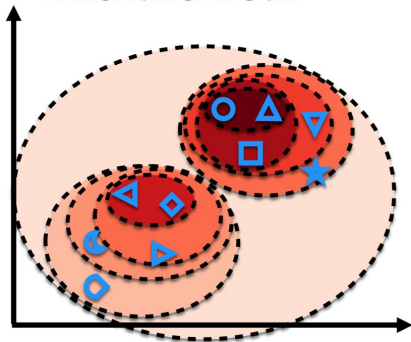
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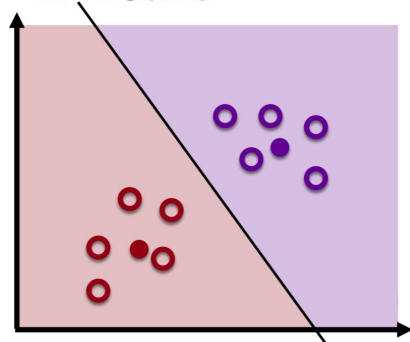


Hierarchical vs K-means

Hierarchical



K-means



How to choose K in K-means?

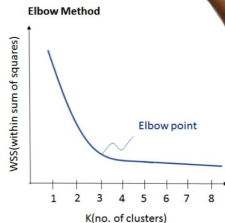
- **Elbow method:** increase K until it does not help to describe data better

How to choose K in K-means?

- **Elbow method:** increase K until it does not help to describe data better
- We are interested in finding K such that the sum of within-group Euclidean distances is smaller

$$J = \sum_{j=1}^K \sum_{i=1}^n \|\mathbf{x}_i^{(j)} - \mathbf{c}_j\|^2,$$

where $\mathbf{c}_j$ is the centroid (mean) of the j^{th} cluster



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- Con:** The number of clusters (K) needs to be defined in advance
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Con: In K Means clustering, the results produced by running the algorithm multiple times might differ because of the random initialization of the centroids. While results are reproducible in Hierarchical clustering.

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4. Medoids of each of the K clusters are updated, taking the ones that are closer to all other points in the cluster

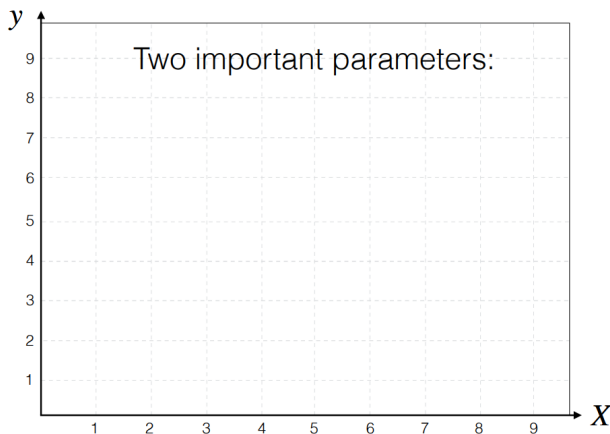
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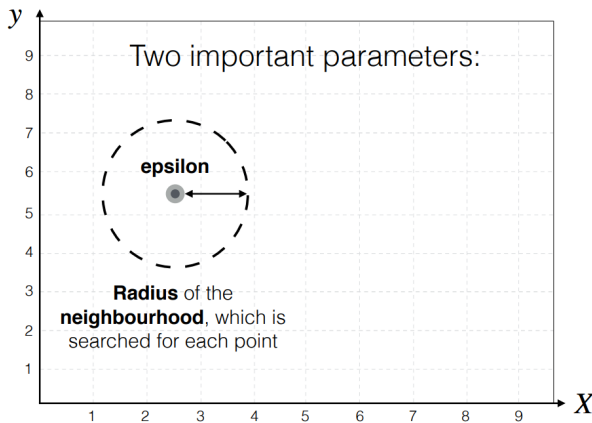
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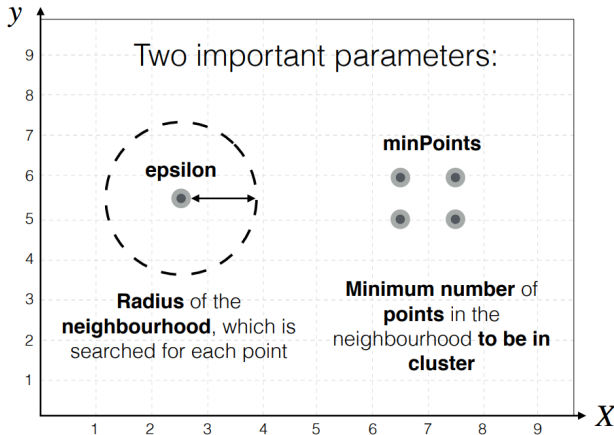
Density-Based Spatial Clustering of Applications with Noise (DBSCAN)



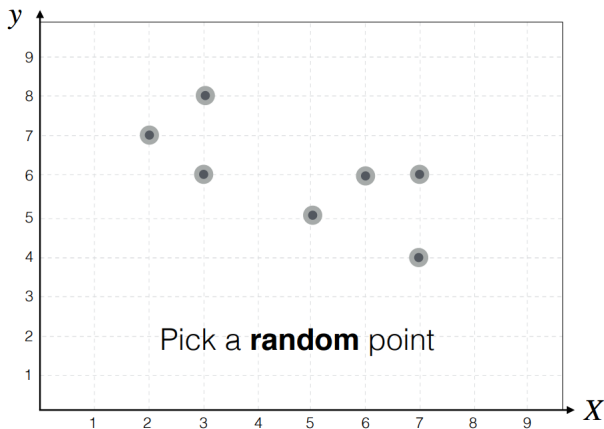
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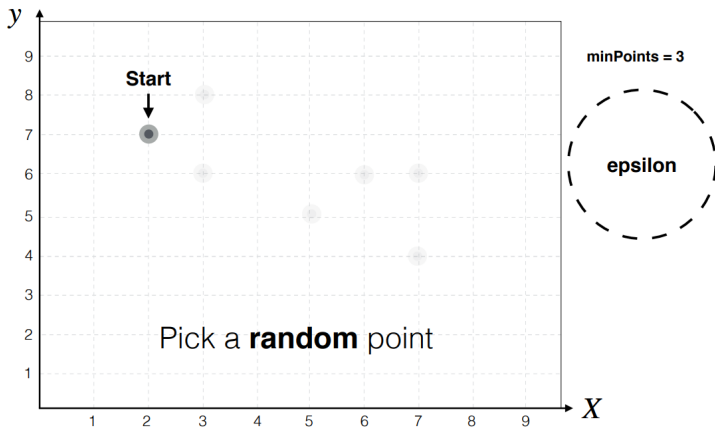
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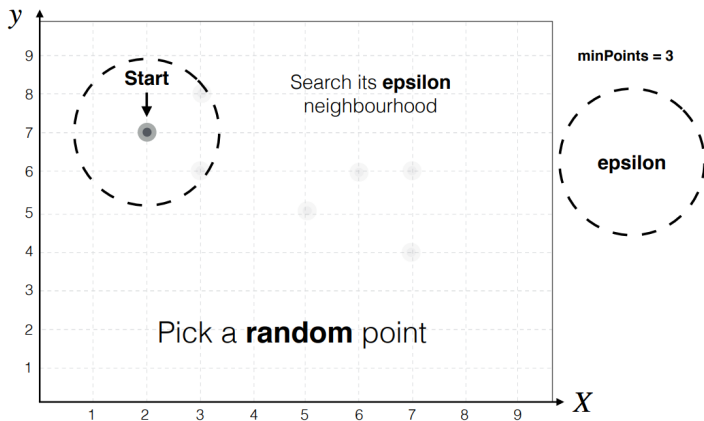
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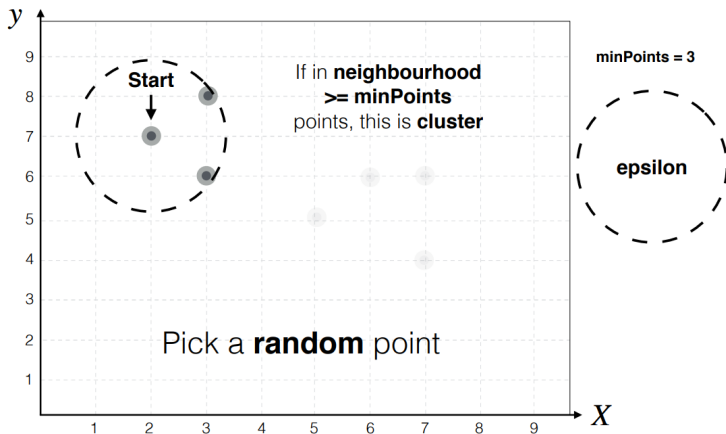
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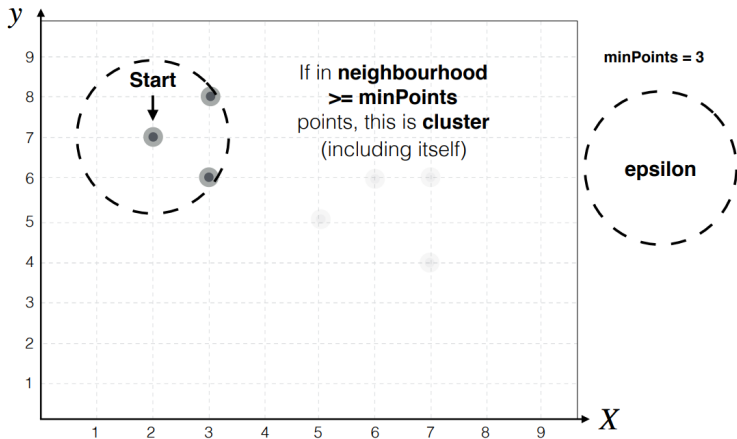
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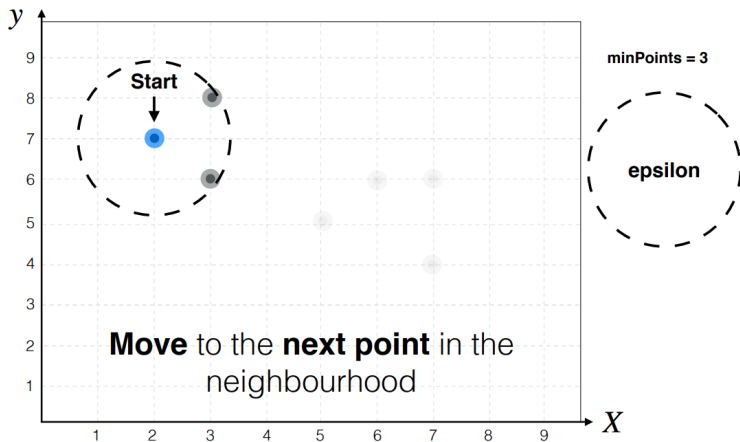
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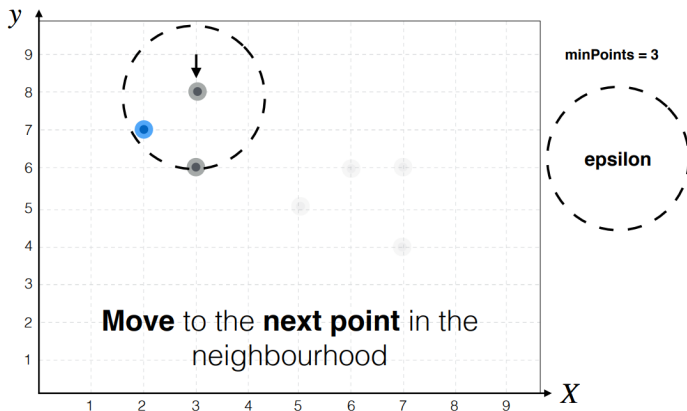
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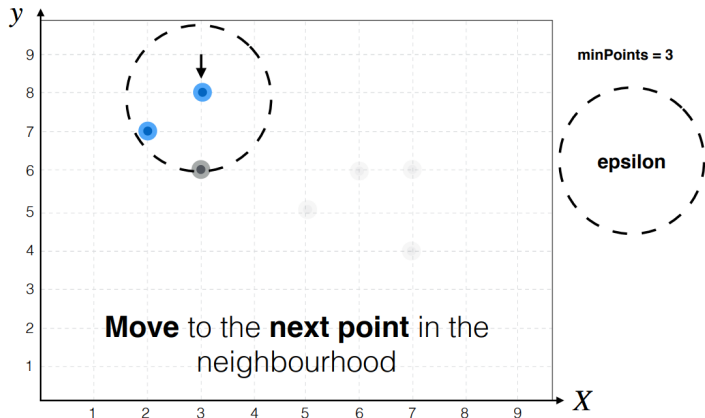
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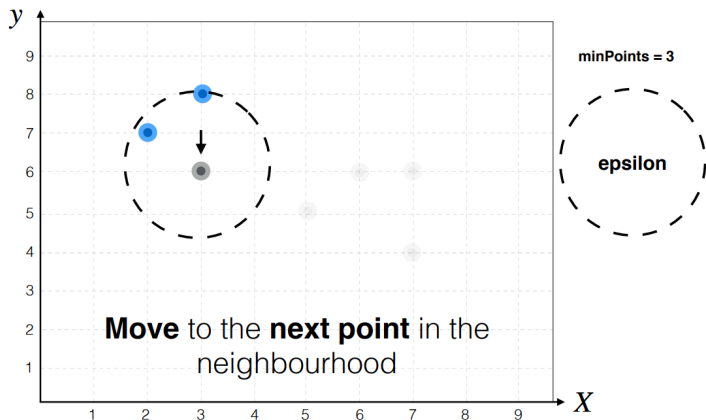
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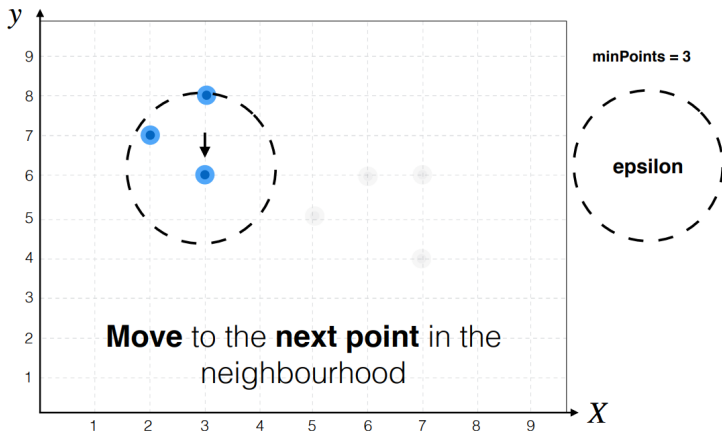
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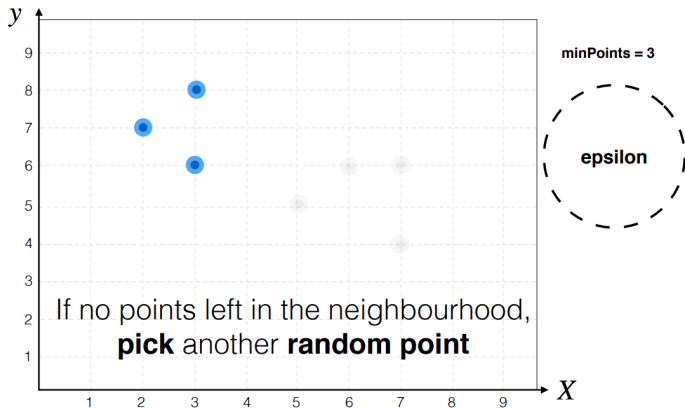
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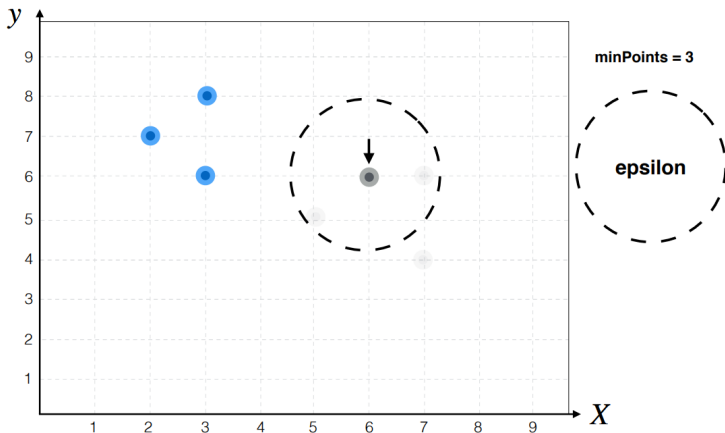
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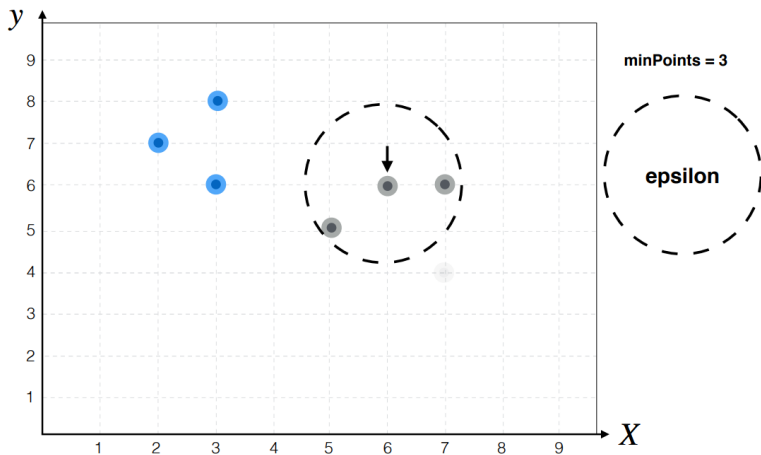
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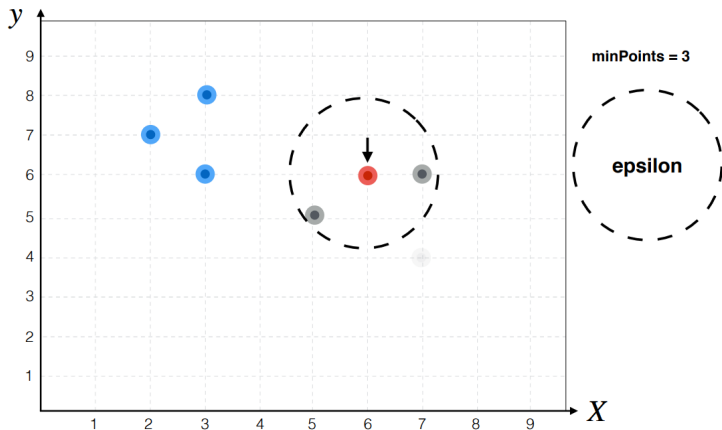
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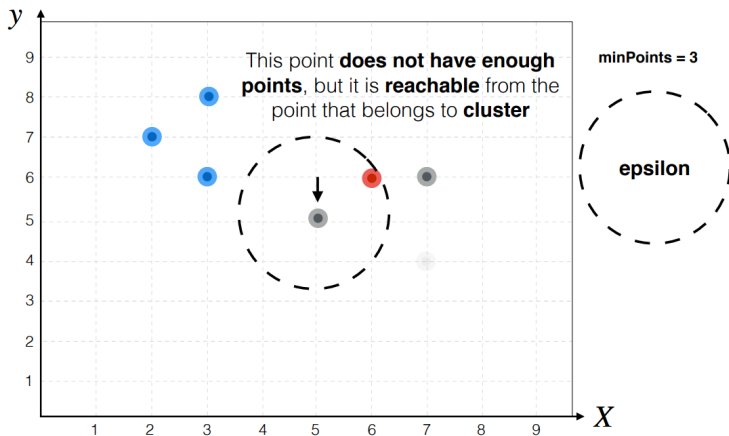
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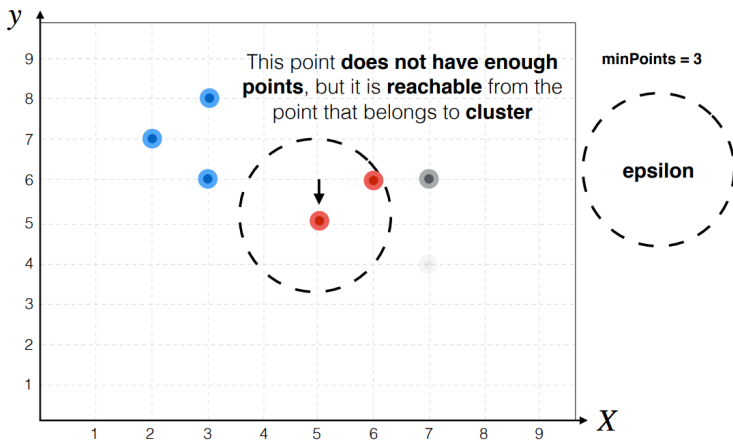
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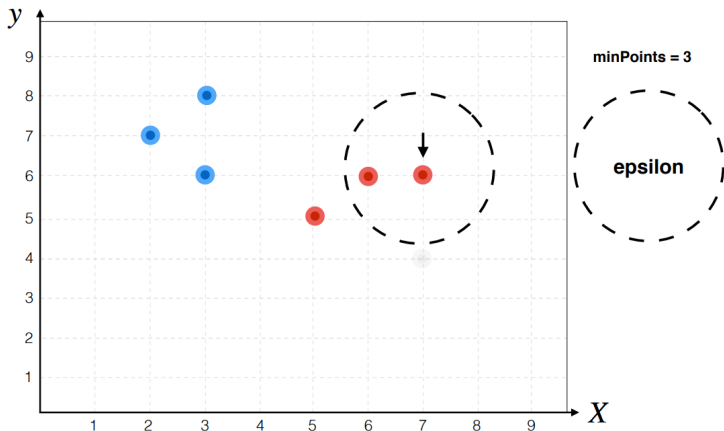
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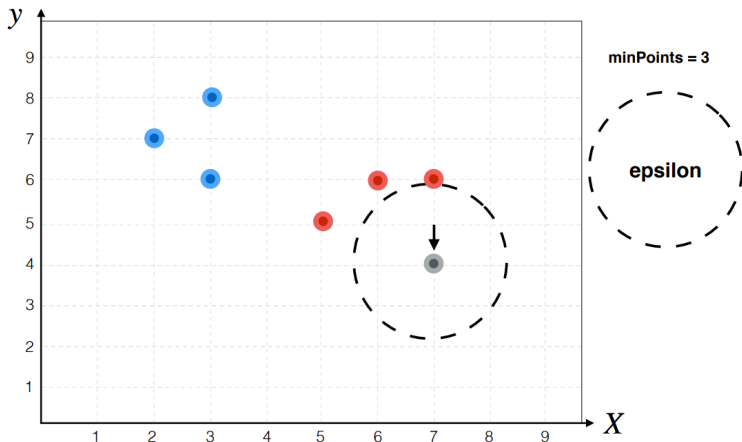
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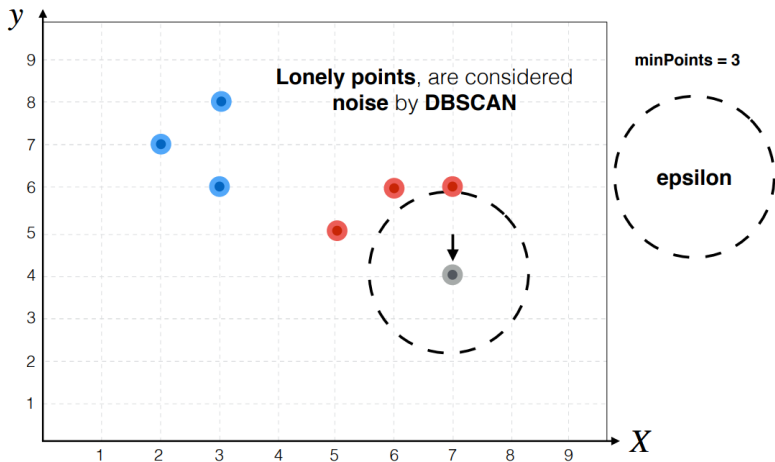
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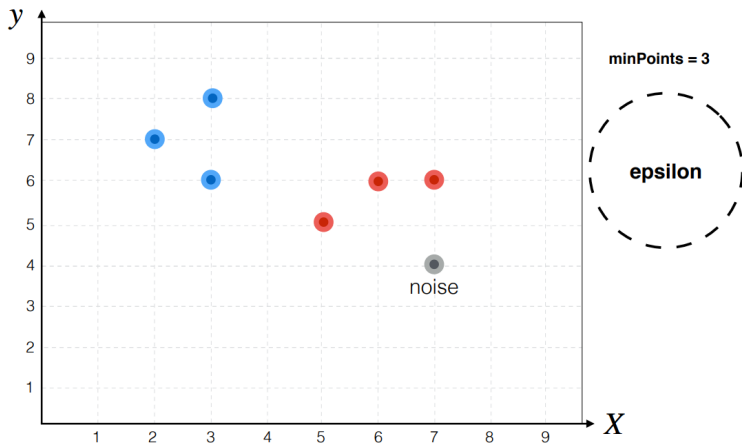
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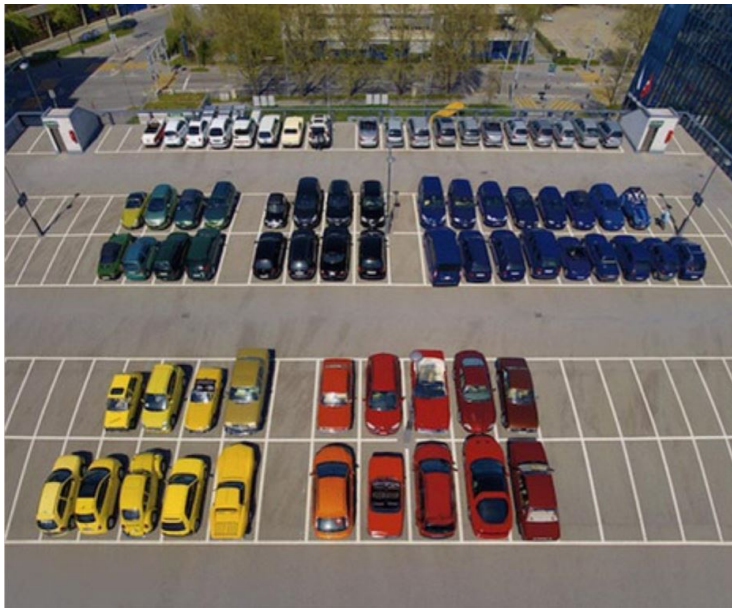
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- On the same dataset the clustering result can be very different if changing the distance measure

Example: Parking Lot



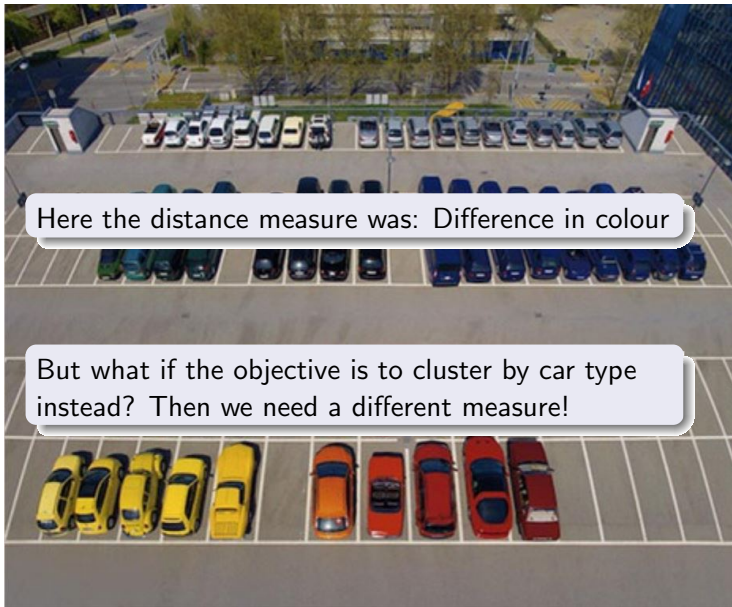
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 - $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)
2. Sometimes triangle inequality is dropped (semi-metric) and we refer to the measure as **dissimilarity** measure

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4. Think about some well-known measures (euclidean, manhattan etc.), see which makes more sense to use or design your own dissimilarity measure

What have we learned today?

- ✓ Concepts of Cluster Analysis
- ✓ Hierarchical Clustering
- ✓ K-means Clustering
- ✓ K-medoids Clustering
- ✓ DBSCAN
- ✓ Dissimilarity Measure Selection